

✓ SOP #6 – ATI (Analysis to Improve) for Cooling Towers

Goal: Enhance tower performance, increase lifespan, minimize water and energy use, and reduce microbial risk

1. Key Optimization Metrics

Metric	Ideal Range or Target	Optimization Action
Condenser ΔT	8–12°F	Clean fill, increase airflow
Blowdown Rate	1–2% of recirculating water	Use automated conductivity controller
Drift Rate	<0.002%	Replace eliminators and repair baffles
Cycles of Concentration	3–7 cycles (treated systems)	Monitor TDS, minimize fresh water makeup
Fan Energy (kW)	<2.5 kW for 100-ton tower	Install VFDs and motor soft starters

2. Strategic Improvements

Install VFDs on Fan Motors: Optimize speed to match real-time demand
 Automate Chemical Dosing & Blowdown: Ensure correct cycles and water safety
 Add Basin Sweep Piping: Reduce sediment and bacteria buildup
 Shade Tower from Direct Sunlight: Lowers algae growth and heat gain
 Seasonal Shutdown Protocol: Full flush, chemical treatment, and dry sealing

3. Legal, Safety & Compliance Considerations

ASHRAE 188: Legionella risk management in cooling towers
 EPA WaterSense Program: Supports efficient water treatment methods
 OSHA 1910 Subpart D & E: Safe access, platforms, and ladder systems
 State/Local Health Codes: Vary by location, often require Legionella testing logs